

# PY1010 — Physics for Biological and Chemical Sciences

Summer 2024 Examination · Paper 1 · Full Marking Scheme

|                           |                                                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Module</b>             | PY1010 — Physics for the Biological and Chemical Sciences                                                                                                                                        |
| <b>Internal examiners</b> | Prof. P. Callanan, Prof. A. Ruth                                                                                                                                                                 |
| <b>Institution</b>        | University College Cork                                                                                                                                                                          |
| <b>Session</b>            | Summer 2024, Paper 1                                                                                                                                                                             |
| <b>Duration</b>           | 180 minutes (+ 15 minutes reading time)                                                                                                                                                          |
| <b>Structure</b>          | Six questions, each given as an either/or choice (A or B). Answer <b>one option</b> of each of the six questions and all parts of that option. Marks for each part are shown in square brackets. |
| <b>Materials</b>          | Non-programmable calculator permitted; constants on the paper's "useful data" page                                                                                                               |

**Coverage note.** This scheme works through **both options (A and B) of all six questions** in full. On the day you only choose one option per question, but working both lets you pick the one you're strongest on.

**Source basis.** Every answer is built **only** from the PY1010 lecture material given to students: the lecture slides (mechanics, fluids, thermodynamics) and the handwritten lecture notes (gas laws, electricity, magnetism, induction, waves and optics). No textbook or outside material is used, except where a highlighted note says the lecture notes don't cover something (Q3(A)(b), (d) and (e)).

**Own-figure rule.** Where a part carries forward a number from an earlier part, an answer that uses the correct method on an earlier wrong value still earns the method marks for the later part.

**Numerical conventions.**  $g = 9.81 \text{ m s}^{-2}$  and other constants from the paper's useful-data page. Intermediate values are kept unrounded, and final answers are given to 2–3 significant figures. Answers differing only through rounding should receive full marks.

Study-Uni

**Question 1 · Option A**

Unit conversion, density, estimating numbers and kinetic energy of red blood cells — 10 marks

Given: Mean RBC volume  $95 \mu\text{m}^3$ ; RBC density  $1125 \text{ kg m}^{-3}$ ; blood volume 4.7 litres;  $4.8 \times 10^6$  RBCs per  $\text{mm}^3$ ; 1 litre =  $10^{-3} \text{ m}^3$ ;  $1 \mu\text{m} = 10^{-6} \text{ m}$ .

**1(A)(a)****[2 marks]**How many  $\mu\text{m}^3$  in  $1 \text{ m}^3$ ?**ANSWER**

$$1 \text{ m}^3 = 10^{18} \mu\text{m}^3$$

**WORKING**

$1 \text{ m} = 10^6 \mu\text{m}$ . Cube both sides (volume has three length dimensions):

$$1 \text{ m}^3 = (10^6 \mu\text{m})^3 = 10^{18} \mu\text{m}^3$$

**EXAM TECHNIQUE**

A very common slip is writing  $10^6 \mu\text{m}^3$ . The conversion factor must be cubed along with the unit:  $(10^6)^3 = 10^{6 \times 3} = 10^{18}$ .

**Marks****Awarded for****1**Uses  $1 \text{ m} = 10^6 \mu\text{m}$  and cubes it.**1** $10^{18}$ .

**1(A)(b)****[2 marks]**

Estimate the mass of a red blood cell.

**ANSWER**

$$m \approx 1.1 \times 10^{-13} \text{ kg}$$

**WORKING**

Convert the volume:  $V = 95 \mu\text{m}^3 = 95 \times 10^{-18} \text{ m}^3$  (using (a)).

Density  $\rho = m/V$ , so  $m = \rho V$ :

$$m = 1125 \text{ kg m}^{-3} \times 95 \times 10^{-18} \text{ m}^3 = 1.07 \times 10^{-13} \text{ kg}$$

**Marks****Awarded for**

**1** Volume converted to  $\text{m}^3$  and  $m = \rho V$  used.

**1** Correct mass with unit.

**1(A)(c)****[3 marks]**

Estimate the total number of red blood cells in an average woman's body.

**ANSWER**

**$$N \approx 2.3 \times 10^{13} \text{ red blood cells}$$**

**WORKING**

Convert the blood volume to  $\text{mm}^3$ .  $1 \text{ m} = 10^3 \text{ mm}$ , so  $1 \text{ m}^3 = 10^9 \text{ mm}^3$ :

$$4.7 \text{ L} = 4.7 \times 10^{-3} \text{ m}^3 = 4.7 \times 10^{-3} \times 10^9 \text{ mm}^3 = 4.7 \times 10^6 \text{ mm}^3$$

Multiply by the number of cells per  $\text{mm}^3$ :

$$N = 4.8 \times 10^6 \text{ mm}^{-3} \times 4.7 \times 10^6 \text{ mm}^3 = 2.3 \times 10^{13}$$

| Marks | Awarded for                                                |
|-------|------------------------------------------------------------|
| 1     | Litres converted to $\text{m}^3$ .                         |
| 1     | $\text{m}^3$ converted to $\text{mm}^3$ ( $\times 10^9$ ). |
| 1     | $N \approx 2.3 \times 10^{13}$ .                           |

**1(A)(d)****[3 marks]**

Estimate the total kinetic energy of all red blood cells when the average velocity of the blood flow is 8 mm/s.

**ANSWER**

$$E_{\text{kin}} \approx 7.7 \times 10^{-5} \text{ J}$$

**WORKING**

Total mass of red blood cells = number  $\times$  mass of one cell:

$$M = 2.256 \times 10^{13} \times 1.069 \times 10^{-13} \text{ kg} = 2.41 \text{ kg}$$

Convert the speed:  $v = 8 \text{ mm/s} = 8 \times 10^{-3} \text{ m s}^{-1}$ . Then:

$$E_{\text{kin}} = \frac{1}{2} M v^2 = \frac{1}{2} \times 2.41 \text{ kg} \times (8 \times 10^{-3} \text{ m s}^{-1})^2 = 7.7 \times 10^{-5} \text{ J}$$

| Marks | Awarded for                                                                  |
|-------|------------------------------------------------------------------------------|
| 1     | Total RBC mass $\approx 2.4 \text{ kg}$ (own-figure from (b), (c) accepted). |
| 1     | Speed in $\text{m s}^{-1}$ and $E_{\text{kin}} = \frac{1}{2}mv^2$ used.      |
| 1     | Correct value $\approx 7.7 \times 10^{-5} \text{ J}$ .                       |

**Question 1(A) Total: 10 Marks**

## Question 1 · Option B

Friction rules, deceleration of a sliding penguin, coefficient of kinetic friction — 10 marks

### 1(B)(a)

[2 marks]

An object rests on a surface. Identify which of the statements (A)–(E) are false, and explain your reasoning.

(A) Frictional forces will always oppose the motion of the object. (B) The frictional force is independent of the normal force. (C) The frictional force is material dependent. (D)  $\mu_s$  is always larger than  $\mu_k$ . (E) The unit of  $\mu_s$  and  $\mu_k$  is  $[\text{N m}^{-1}]$ .

### ANSWER

**(B) and (E) are false. Friction is proportional to the normal force, and  $\mu$  has no unit.**

### WORKING

**(B) is false.** The frictional force is **proportional** to the normal force between the surfaces:  $F_{\text{friction}} \leq \mu_s N$  (at rest),  $F_{\text{friction}} = \mu_k N$  (moving).

**(E) is false.**  $\mu = F_{\text{friction}}/N$  is a force divided by a force, so the coefficients are **unitless**.

(A), (C) and (D) are true: friction always points opposite to the motion, it depends on the materials in contact, and  $\mu_k < \mu_s$ .

| Marks | Awarded for                                               |
|-------|-----------------------------------------------------------|
| 1     | (B) identified as false: friction $\propto$ normal force. |
| 1     | (E) identified as false: $\mu$ is unitless.               |

**1(B)(b)(i)****[3 marks]**

A 12 kg penguin slides on horizontal ice with initial speed  $5.5 \text{ m s}^{-1}$  and comes to rest after 15 m. Calculate the magnitude of its acceleration and show that it is  $\approx 1 \text{ m s}^{-2}$ .

**ANSWER**

$$|a| = 1.01 \text{ m s}^{-2} \approx 1 \text{ m s}^{-2}$$

**WORKING**

Constant acceleration, with  $u = 5.5 \text{ m s}^{-1}$ ,  $v = 0$  and  $s = 15 \text{ m}$ . Use  $v^2 = u^2 + 2as$ :

$$0 = (5.5)^2 + 2a(15) \Rightarrow a = -30.25 \div 30 = -1.01 \text{ m s}^{-2}$$

The negative sign means it is a deceleration. Its magnitude is  $1.01 \approx 1 \text{ m s}^{-2}$ .

| Marks | Awarded for                                     |
|-------|-------------------------------------------------|
| 1     | Correct choice of equation, $v^2 = u^2 + 2as$ . |
| 1     | Correct substitution with $v = 0$ .             |
| 1     | $ a  \approx 1 \text{ m s}^{-2}$ shown.         |

**1(B)(b)(ii)****[2 marks]**

What is the magnitude of the frictional force bringing the penguin to a halt?

**ANSWER**

**$F \approx 12 \text{ N (12.1 N)}$**

**WORKING**

Friction is the only horizontal force, so it is the net force. Newton's second law:

$$F = m a = 12 \text{ kg} \times 1.01 \text{ m s}^{-2} = 12.1 \text{ N}$$

| Marks | Awarded for                                                                                    |
|-------|------------------------------------------------------------------------------------------------|
| 1     | $F = ma$ used.                                                                                 |
| 1     | $F \approx 12 \text{ N}$ (using $a = 1 \text{ m s}^{-2}$ gives exactly 12 N, also full marks). |

**1(B)(b)(iii)****[3 marks]**

What is the coefficient of kinetic friction  $\mu_k$  between the penguin and the ice?

**ANSWER**

**$\mu_k \approx 0.10$  (no unit)**

**WORKING**

On horizontal ice the normal force equals the weight:  $N = mg = 12 \times 9.81 = 117.7 \text{ N}$ .

Kinetic friction  $F = \mu_k N$ , so:

$$\mu_k = F / N = 12.1 \text{ N} \div 117.7 \text{ N} = 0.10$$

| Marks | Awarded for                                                      |
|-------|------------------------------------------------------------------|
| 1     | Normal force $N = mg$ .                                          |
| 1     | $\mu_k = F/N$ .                                                  |
| 1     | $\mu_k \approx 0.10$ , unitless (own-figure from (ii) accepted). |

**Question 1(B) Total: 10 Marks**

## Question 2 · Option A

Mass vs weight, kinematics of Jules Verne's cannon — 10 marks

### 2(A)(a)

[3 marks]

Describe the difference between the mass and weight of an object, giving the SI unit of each.

#### ANSWER

**Mass = amount of matter/inertia, fixed, in kg. Weight = gravitational force  $w = mg$ , depends on  $g$ , in N.**

#### WORKING

**Mass** is the amount of matter in the object and measures its inertia (resistance to changes in motion). It is the same everywhere. SI unit: **kilogram (kg)**.

**Weight** is the gravitational force acting on the object:  $w = mg$ . Because it depends on  $g$ , it changes with location (e.g. smaller on the Moon). SI unit: **newton (N)**.

| Marks | Awarded for                                                     |
|-------|-----------------------------------------------------------------|
| 1     | Mass correctly described (amount of matter / inertia), unit kg. |
| 1     | Weight as a force, $w = mg$ , unit N.                           |
| 1     | Weight depends on $g$ /location, mass does not.                 |

**2(A)(b)(i)****[1 mark]**

The projectile had to reach  $11 \text{ km s}^{-1}$ . Compute the escape velocity in  $\text{m s}^{-1}$ .

**ANSWER**

$$v = 1.1 \times 10^4 \text{ m s}^{-1} \text{ (11 000 m s}^{-1}\text{)}$$

**WORKING**

$$11 \text{ km s}^{-1} \times 1000 \text{ m/km} = 1.1 \times 10^4 \text{ m s}^{-1}$$

**Marks****Awarded for****1**11 000  $\text{m s}^{-1}$ .

**2(A)(b)(ii)****[5 marks]**

If the cannon had a length of 270 m, calculate the acceleration of the passengers whilst in the cannon (constant acceleration, reaching escape velocity as they leave).

**ANSWER**

$$a \approx 2.2 \times 10^5 \text{ m s}^{-2}$$

**WORKING**

Known:  $u = 0$  (starts from rest),  $v = 1.1 \times 10^4 \text{ m s}^{-1}$ ,  $s = 270 \text{ m}$ . Unknown:  $a$ . Time isn't given, so use  $v^2 = u^2 + 2as$ :

$$a = (v^2 - u^2) / (2s) = (1.1 \times 10^4)^2 \div (2 \times 270)$$

$$a = 1.21 \times 10^8 \div 540 = 2.24 \times 10^5 \text{ m s}^{-2}$$

| Marks | Awarded for                                    |
|-------|------------------------------------------------|
| 1     | Knowns identified: $u = 0$ , $v$ , $s$ .       |
| 1     | Correct equation chosen: $v^2 = u^2 + 2as$ .   |
| 1     | Rearranged for $a$ .                           |
| 1     | Correct substitution in SI units.              |
| 1     | $a \approx 2.2 \times 10^5 \text{ m s}^{-2}$ . |

**2(A)(b)(iii)****[1 mark]**

Briefly discuss whether your answer to (ii) is feasible, if the maximum tolerable acceleration is  $50 \times g$  ( $50 \times 9.8 \text{ m s}^{-2}$ ).

**ANSWER**

**Not feasible:  $\approx 23\,000 \text{ g}$ , far above the  $50 \text{ g}$  ( $490 \text{ m s}^{-2}$ ) the body can tolerate**

**WORKING**

$$50 \text{ g} = 50 \times 9.8 = 490 \text{ m s}^{-2}$$

The cannon's acceleration is  $2.24 \times 10^5 \div 9.8 \approx 23\,000 \text{ g}$ , about 460 times the  $50 \text{ g}$  limit.

**Marks****Awarded for****1**Compares with  $490 \text{ m s}^{-2}$  and concludes not feasible.**Question 2(A) Total: 10 Marks**

## Question 2 · Option B

Centripetal acceleration and force in the Bohr model of hydrogen — 10 marks

Given: Electron mass  $9.1 \times 10^{-31} \text{ kg}$ ; orbit radius  $5.3 \times 10^{-11} \text{ m}$ ;  $6.6 \times 10^{15}$  orbits per second.

### 2(B)(a)

[2 marks]

For uniform circular motion write down the expressions for centripetal acceleration and centripetal force.

### ANSWER

$$a_c = v^2/r; F_c = mv^2/r$$

### WORKING

Centripetal acceleration (directed towards the centre of the circle):

$$a_c = v^2 / r ( = \omega^2 r )$$

Centripetal force, from  $F = ma$ :

$$F_c = m v^2 / r ( = m \omega^2 r )$$

| Marks | Awarded for                         |
|-------|-------------------------------------|
| 1     | $a_c = v^2/r$ (or $\omega^2 r$ ).   |
| 1     | $F_c = mv^2/r$ (or $m\omega^2 r$ ). |

**2(B)(b)(i)****[2 marks]**

Calculate the total distance travelled by the electron per orbit.

**ANSWER** **$3.3 \times 10^{-10}$  m per orbit****WORKING**

One orbit is one circumference:

$$C = 2\pi r = 2\pi \times 5.3 \times 10^{-11} \text{ m} = 3.33 \times 10^{-10} \text{ m}$$

| Marks | Awarded for              |
|-------|--------------------------|
| 1     | Circumference $2\pi r$ . |
| 1     | $3.3 \times 10^{-10}$ m. |

**2(B)(b)(ii)****[2 marks]**

Calculate the speed of the electron in its orbit.

**ANSWER** **$v \approx 2.2 \times 10^6 \text{ m s}^{-1}$** **WORKING**Speed = distance per orbit  $\times$  orbits per second ( $v = 2\pi r f$ ):

$$v = 3.33 \times 10^{-10} \text{ m} \times 6.6 \times 10^{15} \text{ s}^{-1} = 2.2 \times 10^6 \text{ m s}^{-1}$$

| Marks | Awarded for                                    |
|-------|------------------------------------------------|
| 1     | $v = 2\pi r \times f$ .                        |
| 1     | $v \approx 2.2 \times 10^6 \text{ m s}^{-1}$ . |

**2(B)(b)(iii)****[2 marks]**

Calculate the centripetal acceleration of the electron.

**ANSWER**

$$a_c \approx 9.1 \times 10^{22} \text{ m s}^{-2}$$

**WORKING**

$$a_c = v^2 / r = (2.198 \times 10^6)^2 \div 5.3 \times 10^{-11} = 9.1 \times 10^{22} \text{ m s}^{-2}$$

| Marks | Awarded for                                                          |
|-------|----------------------------------------------------------------------|
| 1     | $a_c = v^2/r$ with $v$ from (ii).                                    |
| 1     | $\approx 9.1 \times 10^{22} \text{ m s}^{-2}$ (own-figure accepted). |

**2(B)(b)(iv)****[2 marks]**

Calculate the centripetal force of interaction between the electron and proton.

**ANSWER**

$$F_c \approx 8.3 \times 10^{-8} \text{ N}$$

**WORKING**

$$F_c = m a_c = 9.1 \times 10^{-31} \text{ kg} \times 9.1 \times 10^{22} \text{ m s}^{-2} = 8.3 \times 10^{-8} \text{ N}$$

| Marks | Awarded for                              |
|-------|------------------------------------------|
| 1     | $F = ma$ (or $mv^2/r$ ).                 |
| 1     | $\approx 8.3 \times 10^{-8} \text{ N}$ . |

**Question 2(B) Total: 10 Marks**

### Question 3 · Option A

Temperature scales, thermal expansion, density of water — 10 marks

#### 3(A)(a)

[3 marks]

Describe how the Celsius and Absolute (Kelvin) temperature scales are defined, in terms of the ice (freezing) point and steam (boiling) point of water.

#### ANSWER

**Celsius: ice point 0 °C, steam point 100 °C. Kelvin: 273.15 K and 373.15 K, zero at absolute zero.  $T_K = T_{°C} + 273.15$ .**

#### WORKING

**Celsius:** the freezing point of water is defined as **0 °C** and the boiling point as **100 °C**. The range between them is divided into 100 equal degrees.

**Kelvin (absolute):** the freezing point of water is **273.15 K** and the boiling point **373.15 K**. Its zero,  $0\text{ K} = -273.15\text{ °C}$ , is absolute zero: the temperature where the volume of a gas extrapolates to zero.

Only the zero point is shifted between the scales,  $T_K = T_{°C} + 273.15$ . A temperature *difference* is the same on both:  $1\text{ K} = 1\text{ °C}$ .

| Marks | Awarded for                                                       |
|-------|-------------------------------------------------------------------|
| 1     | Celsius fixed points 0 °C and 100 °C.                             |
| 1     | Kelvin fixed points 273.15 K and 373.15 K, zero at absolute zero. |
| 1     | $T_K = T_{°C} + 273.15$ / same size degree.                       |

**3(A)(b)****[1 mark]**

Write down an expression for the linear expansivity of a material, and give the SI unit for the coefficient of expansion.

**ANSWER**

$$\Delta L/L_0 = \alpha \Delta T, \text{ with } \alpha \text{ in } \text{K}^{-1} \text{ (equivalently } ^\circ\text{C}^{-1})$$

**WORKING**

The fractional change in length is proportional to the temperature change:

$$\Delta L / L_0 = \alpha \Delta T$$

$\Delta L/L_0$  has no unit, so  $\alpha$  has units of 1/temperature.

**Note on the lecture material:** The lecture notes give thermal expansivity in its volume form,  $\Delta V/V_0 = \beta \Delta T$  (for a gas). The linear form above is the same relation applied to length.

| Marks | Awarded for |
|-------|-------------|
|-------|-------------|

|   |                                       |
|---|---------------------------------------|
| 1 | Expression and unit $\text{K}^{-1}$ . |
|---|---------------------------------------|

**3(A)(c)****[3 marks]**

The Spire in Dublin is 120 m tall at 20 °C, with  $\alpha = 1.5 \times 10^{-5} \text{ K}^{-1}$ . What is its height on a cold winter day at  $-10 \text{ °C}$ ?

**ANSWER**

**Height  $\approx 119.95 \text{ m}$  (it shrinks by 5.4 cm)**

**WORKING**

Temperature change:  $\Delta T = -10 - 20 = -30 \text{ K}$ .

$$\Delta L = \alpha L_0 \Delta T = 1.5 \times 10^{-5} \times 120 \text{ m} \times (-30) = -0.054 \text{ m}$$

$$L = L_0 + \Delta L = 120 - 0.054 = 119.946 \text{ m}$$

| Marks | Awarded for                                           |
|-------|-------------------------------------------------------|
| 1     | $\Delta T = -30 \text{ K}$ .                          |
| 1     | $\Delta L = \alpha L_0 \Delta T = -0.054 \text{ m}$ . |
| 1     | Final height $\approx 119.95 \text{ m}$ .             |

**3(A)(d)****[2 marks]**

Putting temperature on the horizontal axis, draw a rough graph to show how density varies with temperature for (i) most liquids, (ii) water between 0 °C and 10 °C.

**ANSWER**

**(i) Steadily decreasing line. (ii) Rises to a maximum at 4 °C, then falls.**

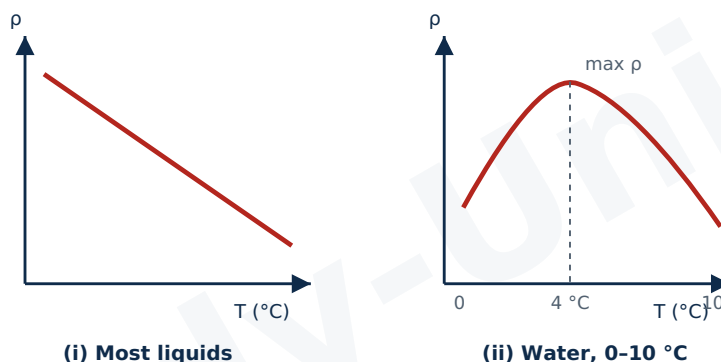
**WORKING**

Fig. 1 — (i) Density falls steadily as temperature rises. (ii) Water's density rises from 0 °C to a maximum at 4 °C, then falls.

(i) Most liquids expand when heated, so their density ( $\rho = m/V$ ) **decreases steadily** as temperature increases.

(ii) Water is unusual. Its density **increases** from 0 °C up to a **maximum at 4 °C**, then decreases as the temperature rises further.

**Note on the lecture material:** Water's density maximum at 4 °C is **not covered in the PY1010 lecture notes**. This answer is included for completeness.

| Marks | Awarded for                                           |
|-------|-------------------------------------------------------|
| 1     | (i) Decreasing density with temperature.              |
| 1     | (ii) Maximum at $\approx 4\text{ }^{\circ}\text{C}$ . |

**3(A)(e)****[1 mark]**

Using the second graph, explain why ponds freeze from the top down and not from the bottom up.

**ANSWER**

**Water below 4 °C is less dense and rises, so the surface reaches 0 °C first and freezes; the densest (4 °C) water stays at the bottom**

**WORKING**

As the pond cools below 4 °C, the colder water becomes **less dense** than the 4 °C water beneath it, so it floats to the top. The coldest water collects at the surface and freezes first. Ice is also less dense than water, so it floats and forms a layer on top, while the denser 4 °C water stays at the bottom.

**Note on the lecture material:** Relies on the same 4 °C behaviour of water, which is not covered in the lecture notes. The idea that less dense material floats does come from the buoyancy lectures.

**Marks****Awarded for****1**

Links the 4 °C density maximum to colder, less dense water rising to the surface.

**Question 3(A) Total: 10 Marks**

### Question 3 · Option B

Buoyancy and Archimedes' principle, a floating block of wood — 10 marks

Given: Wood density  $700 \text{ kg m}^{-3}$ ; volume  $125 \text{ cm}^3$ ; floats in oil with 90% of its volume submerged.

**3(B)(a)**

[2 marks]

Briefly explain the concept of buoyancy and Archimedes' principle.

#### ANSWER

**Buoyant force = weight of fluid displaced ( $F_{\text{buoy}} = \rho_{\text{fl}} V_{\text{disp}} g$ )**

#### WORKING

**Buoyancy:** a fluid exerts an upward force on any object that is fully or partly immersed in it. This happens because the pressure in a fluid increases with depth.

**Archimedes' principle:** the upward (buoyant) force equals the **weight of the fluid displaced** by the body:  $F_{\text{buoy}} = \rho_{\text{fluid}} V_{\text{displaced}} g$ . An object floats if the buoyant force can balance its weight, and sinks if it cannot.

| Marks | Awarded for                                                  |
|-------|--------------------------------------------------------------|
| 1     | Buoyancy as an upward force from the fluid.                  |
| 1     | Archimedes' principle: equals the weight of fluid displaced. |

**3(B)(b)(i)****[2 marks]**

Calculate the mass of the wood.

**ANSWER**

**$m = 0.0875 \text{ kg (87.5 g)}$**

**WORKING**Convert the volume:  $125 \text{ cm}^3 = 125 \times 10^{-6} \text{ m}^3$ .

$$m = \rho V = 700 \text{ kg m}^{-3} \times 1.25 \times 10^{-4} \text{ m}^3 = 0.0875 \text{ kg}$$

| Marks | Awarded for                             |
|-------|-----------------------------------------|
| 1     | Volume in $\text{m}^3$ , $m = \rho V$ . |
| 1     | 0.0875 kg.                              |

**3(B)(b)(ii)****[4 marks]**

Calculate the density of the oil.

**ANSWER**

$$\rho_{\text{oil}} \approx 7.8 \times 10^2 \text{ kg m}^{-3} \text{ (778 kg m}^{-3}\text{)}$$

**WORKING**

The block floats in equilibrium, so the buoyant force equals its weight:

$$\rho_{\text{oil}} \times (0.9 V) \times g = m g = \rho_{\text{wood}} V g$$

V and g cancel:

$$\rho_{\text{oil}} = \rho_{\text{wood}} / 0.9 = 700 \div 0.9 = 778 \text{ kg m}^{-3}$$

| Marks | Awarded for                          |
|-------|--------------------------------------|
| 1     | Floating: buoyant force = weight.    |
| 1     | Displaced volume = 0.9V.             |
| 1     | Rearranged for $\rho_{\text{oil}}$ . |
| 1     | $\approx 778 \text{ kg m}^{-3}$ .    |

**3(B)(b)(iii)****[2 marks]**

When you pour the oil into water the two do not mix. Would you expect the oil to be above or below the water? Briefly justify.

**ANSWER**

**Above the water, because oil is less dense ( $778 < 1000 \text{ kg m}^{-3}$ )**

**WORKING**

The oil ( $\approx 778 \text{ kg m}^{-3}$ ) is **less dense** than water ( $1000 \text{ kg m}^{-3}$ ). A less dense substance displaces a weight of water greater than its own weight, so the buoyant force pushes it up and it floats.

| Marks | Awarded for                                  |
|-------|----------------------------------------------|
| 1     | Above.                                       |
| 1     | Justified by comparing densities / buoyancy. |

**Question 3(B) Total: 10 Marks**

## Question 4 · Option A

Frequency, types of wave,  $v = f\lambda$ , seismic wave — 10 marks

### 4(A)(a)

[2 marks]

What is meant by the frequency of a wave? Give its SI unit.

#### ANSWER

**$f$  = number of oscillations per second ( $f = 1/P$ ); unit hertz,  $\text{Hz} = \text{s}^{-1}$**

#### WORKING

The frequency  $f$  is the number of complete oscillations (crests passing a point) per second. It is the inverse of the period  $P$ , the time for one oscillation:  $f = 1/P$ .

| Marks | Awarded for                                      |
|-------|--------------------------------------------------|
| 1     | Definition (oscillations per second / 1/period). |
| 1     | Hz ( $\text{s}^{-1}$ ).                          |

4(A)(b)

[4 marks]

Describe the two main types of wave. Give four examples of wave motion.

ANSWER

**Transverse ( $\perp$  to propagation) and longitudinal ( $\parallel$  to propagation); examples: water, seismic, sound, light**

WORKING

**Transverse waves:** the medium moves **perpendicular** to the direction the wave travels, e.g. waves on a rope, water waves, light.

**Longitudinal waves:** the medium vibrates **parallel** to the direction the wave travels, forming compressions and rarefactions, e.g. sound waves in air.

**Four examples of wave motion:** water waves, seismic waves, sound, light.

| Marks | Awarded for                        |
|-------|------------------------------------|
| 1     | Transverse described correctly.    |
| 1     | Longitudinal described correctly.  |
| 2     | Four valid examples (½ mark each). |

**4(A)(c)****[1 mark]**

Write down the expression relating the velocity, frequency and wavelength of a wave.

**ANSWER**

$$v = f\lambda$$

**WORKING**

$$v = f \lambda$$

$v$  in  $\text{m s}^{-1}$ ,  $f$  in Hz,  $\lambda$  in m. This holds for all waves.

**Marks****Awarded for****1** $v = f\lambda$ .

**4(A)(d)****[3 marks]**

A seismic wave has frequency 10 Hz and wavelength 500 m. If it took 100 s to travel from the earthquake to the geologist, how far away was the earthquake?

**ANSWER**

**$d = 5 \times 10^5 \text{ m (500 km)}$**

**WORKING**

$$v = f \lambda = 10 \text{ Hz} \times 500 \text{ m} = 5000 \text{ m s}^{-1}$$

It travels in a straight line at constant speed, so distance = speed  $\times$  time:

$$d = v t = 5000 \text{ m s}^{-1} \times 100 \text{ s} = 5.0 \times 10^5 \text{ m}$$

| Marks | Awarded for                              |
|-------|------------------------------------------|
| 1     | $v = f\lambda = 5000 \text{ m s}^{-1}$ . |
| 1     | $d = vt$ .                               |
| 1     | 500 km.                                  |

**Question 4(A) Total: 10 Marks**

## Question 4 · Option B

Resistance, current and power; charge flowing through a resistor — 10 marks

Given: A charge of 2 C flows through a 10  $\Omega$  resistance in 5 s.

**4(B)(a)**

**[3 marks]**

Define (i) electric resistance, (ii) electric current, and (iii) electric power.

### ANSWER

$$R = V/I \text{ (}\Omega\text{)}; I = Q/t \text{ (A)}; P = E/t = IV \text{ (W)}$$

### WORKING

(i) **Resistance** is the ratio of the potential difference across a conductor to the current through it,  $R = V/I$ . It measures how strongly the conductor opposes the flow of charge. Unit: ohm ( $\Omega$ ).

(ii) **Current** is the rate of flow of charge,  $I = Q/t$ . Unit: ampere,  $A = C s^{-1}$ .

(iii) **Power** is the rate at which electrical energy is used (work done),  $P = E/t = IV$ . Unit: watt (W).

| Marks | Awarded for                         |
|-------|-------------------------------------|
| 1     | Resistance $R = V/I$ .              |
| 1     | Current $I = Q/t$ .                 |
| 1     | Power = energy per time, $P = IV$ . |

**4(B)(b)(i)****[1 mark]**

What is the current through the resistance?

**ANSWER**

**$I = 0.4 \text{ A}$**

**WORKING**

$$I = Q / t = 2 \text{ C} \div 5 \text{ s} = 0.4 \text{ A}$$

**Marks****Awarded for****1**

0.4 A.

**4(B)(b)(ii)****[2 marks]**

How much work is done on the charge?

**ANSWER**

**$W = 8 \text{ J}$**

**WORKING**Voltage across the resistor:  $V = IR = 0.4 \text{ A} \times 10 \Omega = 4 \text{ V}$ .Work done moving charge  $q$  through potential difference  $V$  is  $W = Vq$ :

$$W = V q = 4 \text{ V} \times 2 \text{ C} = 8 \text{ J}$$

**Marks****Awarded for****1** $V = IR = 4 \text{ V}$ .**1** $W = Vq = 8 \text{ J}$ .

**4(B)(b)(iii)****[2 marks]**

At what rate is power dissipated by the resistor?

**ANSWER**

**$P = 1.6 \text{ W}$**

**WORKING**

$$P = W / t = 8 \text{ J} \div 5 \text{ s} = 1.6 \text{ W}$$

Check:  $P = I^2 R = (0.4)^2 \times 10 = 1.6 \text{ W}$ .

| Marks    | Awarded for                  |
|----------|------------------------------|
| <b>1</b> | $P = W/t$ (or $I^2 R$ , IV). |
| <b>1</b> | 1.6 W.                       |

**4(B)(b)(iv)****[2 marks]**

A second, identical resistor is connected in series with the first. What is the total voltage drop across both resistors if the current stays the same?

**ANSWER**

$$V_{\text{total}} = 8 \text{ V}$$

**WORKING**

In series, the same 0.4 A flows through both resistors. Each one has  $V = IR = 0.4 \times 10 = 4 \text{ V}$  across it.

V is the work done per unit charge, so the voltage drops across the two resistors add:

$$V_{\text{total}} = 4 \text{ V} + 4 \text{ V} = 8 \text{ V} ( = I \times 20 \Omega )$$

| Marks | Awarded for                                                                      |
|-------|----------------------------------------------------------------------------------|
| 1     | Same current through both; 4 V across each (or $R_{\text{total}} = 20 \Omega$ ). |
| 1     | 8 V.                                                                             |

**Question 4(B) Total: 10 Marks**

## Question 5 · Option A

Lorentz force on a wire, the electric motor loop, field of a current loop — 10 marks

### 5(A)(a)

[4 marks]

Write down the formula for the magnitude of the Lorentz force on a current-carrying wire in an external field  $B$ , with the current perpendicular to  $B$ . Draw a diagram relating the directions of the current,  $B$  and the Lorentz force.

### ANSWER

**$F = B I \ell$ , with  $F$  perpendicular to both  $I$  and  $B$  (Left Hand Rule)**

### WORKING

$$F = B I \ell$$

$F$  is the force (N),  $B$  the magnetic field (T),  $I$  the current (A) and  $\ell$  the length of wire in the field (m).

The force is perpendicular to **both**  $I$  and  $B$ . Use the Left Hand Rule: first finger along  $B$ , second finger along  $I$ , and the thumb gives  $F$ .

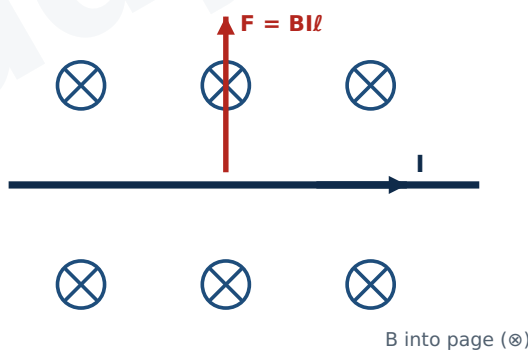


Fig. 2 — Current to the right in a field into the page: the Lorentz force points up the page, perpendicular to both  $I$  and  $B$ .

| Marks | Awarded for                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------|
| 1     | $F = B I \ell$ .                                                                                    |
| 1     | Terms defined ( $B$ , $I$ , $\ell$ ).                                                               |
| 2     | Diagram with $I$ , $B$ and $F$ mutually perpendicular in a consistent (left-hand-rule) arrangement. |

**5(A)(b)(i)****[2 marks]**

The loop's plane is parallel to B (as shown) and I flows in the direction shown. Does the loop rotate (A) clockwise, (B) anticlockwise, or (C) remain fixed?

**ANSWER****(B) Anticlockwise****WORKING**

B points from the N pole to the S pole (left → right). The two sides of the loop that run along B feel no force (I parallel to B). Only the two sides that run along the axis direction, perpendicular to B, feel a force.

Following the current from the lead wires: on the **left** side it flows **into** the page (away from you), and on the **right** side it flows **out of** the page (towards you).

Left Hand Rule, with B to the right: the left side is pushed **down** and the right side is pushed **up**. The two forces are equal and opposite on either side of the axis, so they turn the loop (a torque).

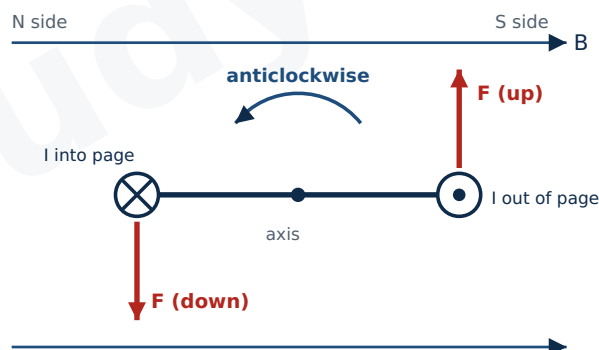


Fig. 3 — View along the axis of rotation: the left side is pushed down and the right side up, so the loop turns anticlockwise.

| Marks | Awarded for                                                                    |
|-------|--------------------------------------------------------------------------------|
| 1     | Forces only on the sides perpendicular to B, in opposite directions (up/down). |
| 1     | (B) anticlockwise.                                                             |

**5(A)(b)(ii)****[2 marks]**

If I flows in the opposite direction around the loop and the direction of B is reversed, which of (A), (B), (C) now holds?

**ANSWER****(B) Anticlockwise, the same as before****WORKING**

Reversing I reverses the force on each side. Reversing B also reverses the force on each side. Both reversals together cancel, so each side is pushed the same way as in (i).

| Marks | Awarded for                               |
|-------|-------------------------------------------|
| 1     | Recognises that the two reversals cancel. |
| 1     | (B) anticlockwise.                        |

**5(A)(c)****[2 marks]**

Consider the magnetic field generated by the current in a small circular loop of wire (no external field). Sketch the configuration of this field, and relate it to the magnetic field of one other type of magnetic object.

**ANSWER**

**A dipole field, like that of a bar magnet (N on one face of the loop, S on the other)**

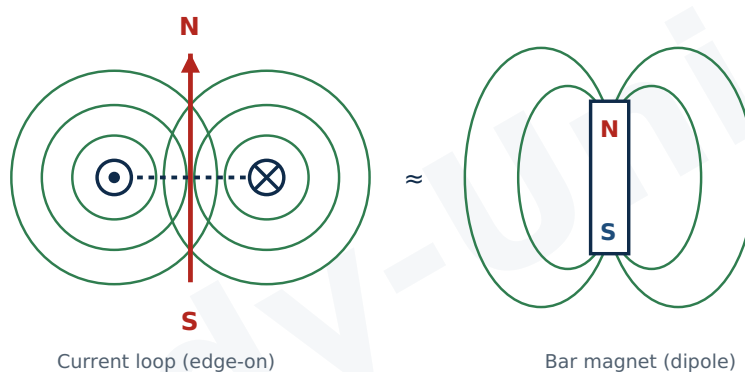
**WORKING**

Fig. 4 — Field lines circle each side of the loop (right-hand rule) and all pass the same way through its centre, giving a dipole field like a bar magnet.

Using the right-hand rule, the field lines circle around each side of the loop. They all pass through the centre of the loop in the **same direction**, then loop back around the outside.

This is a **dipole** field, the same shape as the field of a **bar magnet**. One face of the loop acts as a north pole and the other as a south pole. (The same idea explains the Earth's magnetic field and magnetism in atoms.)

| Marks | Awarded for                                                            |
|-------|------------------------------------------------------------------------|
| 1     | Sketch: field lines through the centre of the loop, returning outside. |
| 1     | Related to a bar magnet (or Earth / dipole).                           |

**Question 5(A) Total: 10 Marks**

## Question 5 · Option B

Faraday's law of induction, magnetic flux, induced voltage and current — 10 marks

Given: Circular loop of area  $0.3 \text{ m}^2$ , held horizontal; a vertical  $B$  increases from 0 to 2 T in 3 s.

### 5(B)(a)

[1 mark]

Write down Faraday's Law of induction, explaining each of the terms present.

#### ANSWER

**$V = \Delta\Phi/\Delta t$ : induced voltage = rate of change of magnetic flux**

#### WORKING

$$V = \Delta\Phi / \Delta t$$

$V$  is the induced voltage (V).  $\Delta\Phi$  is the change in magnetic flux through the circuit (Wb).  $\Delta t$  is the time taken for the change (s). A voltage is induced only while the flux is changing.

Marks

Awarded for

1

Law with terms explained.

### 5(B)(b)

[1 mark]

Give the SI unit for magnetic flux.

#### ANSWER

**The weber, Wb (= T m<sup>2</sup>)**

#### WORKING

Magnetic flux is  $\Phi = BA$ , so its unit is T m<sup>2</sup>. This is called the weber.

Marks

Awarded for

1

Wb (or T m<sup>2</sup>).

**5(B)(c)(i)****[2 marks]**

Between  $t = 0$  and  $t = 3$  s, what is the increase in magnetic flux through the loop?

**ANSWER**

$$\Delta\Phi = 0.6 \text{ Wb}$$

**WORKING**

B is vertical and the loop is horizontal, so B is perpendicular to the loop and  $\Phi = BA$ :

$$\Delta\Phi = \Delta B \times A = (2 \text{ T} - 0) \times 0.3 \text{ m}^2 = 0.6 \text{ Wb}$$

| Marks | Awarded for                      |
|-------|----------------------------------|
| 1     | $\Phi = BA$ with $B \perp$ loop. |
| 1     | 0.6 Wb.                          |

**5(B)(c)(ii)****[2 marks]**

What is the induced voltage?

**ANSWER**

$$V = 0.2 \text{ V}$$

**WORKING**

$$V = \Delta\Phi / \Delta t = 0.6 \text{ Wb} \div 3 \text{ s} = 0.2 \text{ V}$$

| Marks | Awarded for            |
|-------|------------------------|
| 1     | Faraday's law applied. |
| 1     | 0.2 V.                 |

**5(B)(c)(iii)****[2 marks]**

If the resistance of the wire is  $2\ \Omega$ , what current flows?

**ANSWER**

$$I = 0.1\text{ A}$$

**WORKING**

$$I = V / R = 0.2\text{ V} \div 2\ \Omega = 0.1\text{ A}$$

| Marks | Awarded for |
|-------|-------------|
| 1     | $I = V/R$ . |
| 1     | 0.1 A.      |

**5(B)(c)(iv)****[2 marks]**

For times greater than 3 s, B stays fixed at 2 T. Does the induced current continue to increase, stay the same, or drop to zero?

**ANSWER**

**It drops to zero: the flux is no longer changing**

**WORKING**

Once B stops changing, the flux through the loop is constant, so  $\Delta\Phi/\Delta t = 0$ . A voltage and current are induced only **while** the flux is changing, so there is no induced voltage and no current.

| Marks | Awarded for                                 |
|-------|---------------------------------------------|
| 1     | Drops to zero.                              |
| 1     | Reason: no change in flux, so no induced V. |

**Question 5(B) Total: 10 Marks**

**Question 6 · Option A**

Lens sign conventions, magnification, the thin lens equation — 10 marks

**6(A)(a)(i)****[1 mark]**

The focal length  $f$  of a concave lens is (A) always positive, (B) always negative, or (C) can be either.

**ANSWER****(B) Always negative****WORKING**

With the sign convention used for the thin lens equation,  $f < 0$  for a concave (diverging) lens, always.

| Marks | Awarded for |
|-------|-------------|
|-------|-------------|

|   |      |
|---|------|
| 1 | (B). |
|---|------|

**6(A)(a)(ii)****[1 mark]**

The focal length  $f$  of a convex lens is (A) always positive, (B) always negative, or (C) can be either.

**ANSWER****(A) Always positive****WORKING**

$f > 0$  for a convex (converging) lens, always.

| Marks | Awarded for |
|-------|-------------|
|-------|-------------|

|   |      |
|---|------|
| 1 | (A). |
|---|------|

**6(A)(a)(iii)****[1 mark]**

The image distance  $v$  of a convex lens is (A) always positive, (B) always negative, or (C) can be either.

**ANSWER****(C) Can be either****WORKING**

If the object is further from the lens than  $f$ , the image is real and  $v > 0$ . If the object is closer than  $f$  ( $u < f$ , a magnifying glass), the image is virtual and  $v < 0$ . For example,  $u = 10$  cm and  $f = 20$  cm give  $v = -20$  cm.

| Marks | Awarded for |
|-------|-------------|
| 1     | (C).        |

**6(A)(a)(iv)****[1 mark]**

For object and image distances  $u$  and  $v$ , the magnification of a convex lens is (A) always greater than one, (B) equals  $v/u$ , or (C) both of these.

**ANSWER****(B) Equals  $v/u$** **WORKING**

From similar triangles,  $M = h_i/h_o = v/u$ . It is **not** always greater than one: for  $u = 10$  cm and  $f = 4$  cm,  $M = 0.66$ , so the image is reduced in size. So only (B) holds.

| Marks | Awarded for |
|-------|-------------|
| 1     | (B).        |

**6(A)(b)(i)****[1 mark]**

A converging lens ( $f = 30 \text{ cm}$ ) forms a real image three times further from the lens than the object. Is the image erect or inverted?

**ANSWER****Inverted****WORKING**

A real image from a converging lens forms where the rays actually cross, on the opposite side of the lens. Ray tracing shows this image is **inverted**.

| Marks | Awarded for |
|-------|-------------|
| 1     | Inverted.   |

**6(A)(b)(ii)****[2 marks]**

What is the magnification of the object?

**ANSWER****M = 3 (image three times the size of the object, inverted)****WORKING**

The image distance is three times the object distance,  $v = 3u$ :

$$M = v / u = 3u / u = 3$$

| Marks | Awarded for |
|-------|-------------|
| 1     | $M = v/u$ . |
| 1     | $M = 3$ .   |

**6(A)(b)(iii)****[3 marks]**

What are the distances of the object and the image from the lens?

**ANSWER****Object at  $u = 40$  cm; image at  $v = 120$  cm****WORKING**

Thin lens equation with  $f = +30$  cm (convex) and  $v = 3u$  (real image,  $v > 0$ ):

$$1/u + 1/(3u) = 1/30 \Rightarrow 4/(3u) = 1/30 \Rightarrow u = 40 \text{ cm}$$

$$v = 3u = 120 \text{ cm}$$

| Marks | Awarded for                        |
|-------|------------------------------------|
| 1     | Thin lens equation with $v = 3u$ . |
| 1     | $u = 40$ cm.                       |
| 1     | $v = 120$ cm.                      |

**Question 6(A) Total: 10 Marks**

## Question 6 · Option B

Snell's law, refraction at the water surface seen by a diver — 10 marks

Given: A submerged diver sees the Sun  $35^\circ$  from the vertical;  $n_{\text{water}} = 1.33$ .

### 6(B)(a)

[3 marks]

Write down Snell's Law of Refraction and, with the aid of a diagram, briefly explain all the terms present.

### ANSWER

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \text{ (angles measured from the normal)}$$

### WORKING

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

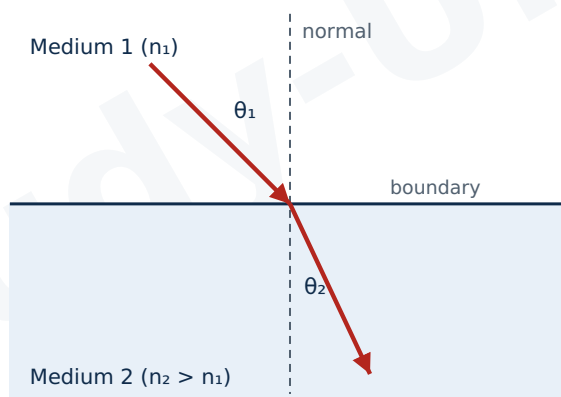


Fig. 5 — A ray passing into an optically denser medium bends towards the normal.

$n_1$ ,  $n_2$ : refractive indices of media 1 and 2,  $n = c /$  (speed of light in the medium). For example, air  $\approx 1$  and water  $\approx 1.33$ .

$\theta_1$ : angle of incidence and  $\theta_2$ : angle of refraction, both measured from the **normal** (the line perpendicular to the boundary). The incident ray, refracted ray and normal all lie in the same plane.

| Marks | Awarded for                                                |
|-------|------------------------------------------------------------|
| 1     | Snell's law stated.                                        |
| 1     | $n$ and $\theta$ terms explained (angles from the normal). |
| 1     | Labelled diagram.                                          |

**6(B)(b)(i)****[3 marks]**

Draw a diagram of the path of a ray of light from the Sun, noting the angles of incidence as observed by the diver and her friend on the shore.

**ANSWER**

**Ray bends towards the vertical on entering the water;  $\theta_{\text{air}} > \theta_{\text{water}} = 35^\circ$**

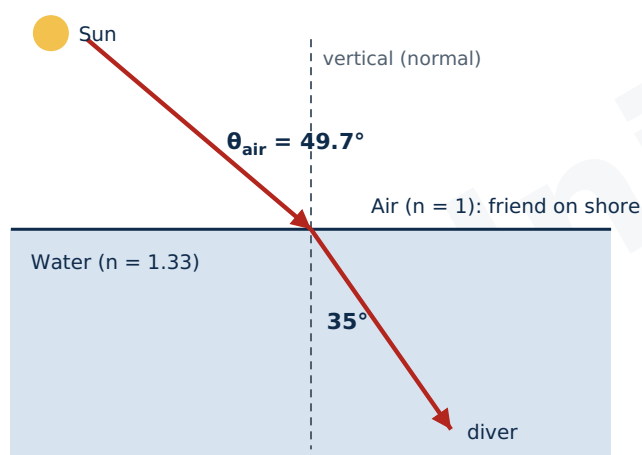
**WORKING**

Fig. 6 — The Sun's ray bends towards the normal as it enters the water:  $49.7^\circ$  from the vertical in air (friend) becomes  $35^\circ$  in water (diver).

The water surface is horizontal, so the **normal is vertical** and angles "from the vertical" are the angles to the normal.

In air (seen by the friend) the ray makes the larger angle,  $\theta_{\text{air}}$ . On entering the denser water it bends **towards** the normal, to  $35^\circ$  (seen by the diver).

| Marks | Awarded for                                                             |
|-------|-------------------------------------------------------------------------|
| 1     | Horizontal surface with vertical normal drawn.                          |
| 1     | Ray bending towards the normal in the water.                            |
| 1     | Angles labelled: $\theta_{\text{air}}$ (friend) and $35^\circ$ (diver). |

**6(B)(b)(ii)****[4 marks]**

At what angle from the vertical does the diver's friend see the Sun? ( $n_{\text{water}} = 1.33$ )

**ANSWER**

**The friend sees the Sun  $\approx 50^\circ$  (49.7°) from the vertical**

**WORKING**

Snell's law at the water surface, with air  $n = 1$  and water  $n = 1.33$ :

$$1 \times \sin \theta_{\text{air}} = 1.33 \times \sin 35^\circ$$

$$\sin \theta_{\text{air}} = 1.33 \times 0.5736 = 0.763$$

$$\theta_{\text{air}} = \sin^{-1}(0.763) = 49.7^\circ$$

| Marks | Awarded for                                |
|-------|--------------------------------------------|
| 1     | Snell's law set up with the correct media. |
| 1     | $35^\circ$ assigned to the water side.     |
| 1     | $\sin \theta_{\text{air}} = 0.763$ .       |
| 1     | $\theta_{\text{air}} \approx 49.7^\circ$ . |

**Question 6(B) Total: 10 Marks**